

The Brightest in Space

The brightest of all time?

Aaryaman Dahiya |
feedback@digit.in

P

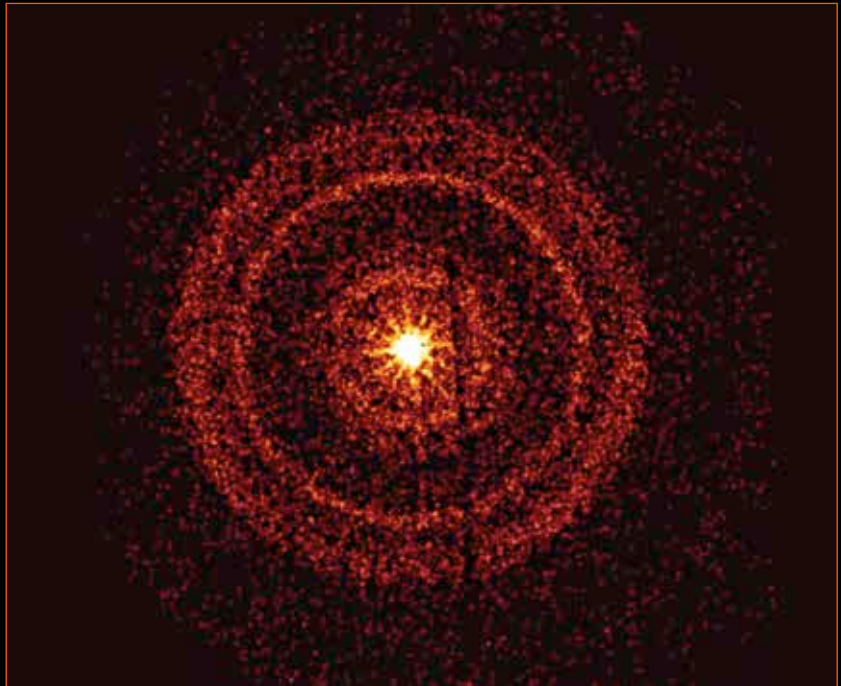
reviously, the Fermi Gamma-ray Space Telescope had detected a Short Gamma-ray burst

(GRB) that would have lasted only a second. It was initially believed that such bursts come from neutron star smashups but someone who closely inspected it concluded that it possibly came from the implosion of a massive star. However, this process is usually believed to cause a GRB that is longer than two seconds but in this case, that was not the case. This raised a lot of questions around GRBs, what causes them and their impact. Let's learn more about this.

WHAT REALLY HAPPENED?

On October 9 at around 10:15 AM Eastern Time, NASA's gamma-ray telescope which is known as the Neil Gehrels Swift Observatory in space alerted astronomers that something peculiar might be taking place. A gamma ray burst was not what came to mind first for the astronomers. They suspected that it was within the Milky Way itself, but something like that happening in our own galaxy would be visible to anyone, however, that was not the case.

It was soon learnt that the Fermi Gamma-ray, which is again



A photo of BOAT taken an hour after the burst was detected

Image Source: ScienceNews.org

a space telescope belonging to NASA, had captured the flash and it was probably the brightest one it had ever recorded. After 50 years of observing these explosions, it was confirmed that this was the brightest one ever witnessed in history. But what does it really mean to witness such a thing and what is the noise all about?



WHY IS IT BEING SPOKEN ABOUT SO MUCH?

But let us answer what Gamma-ray Burst (GRB) is in the first place? Firstly there are two types of GRBs. A Long GRB appears when there is the death of a massive star and it leaves behind itself a Neutron Star or a Black Hole. On the other hand a Short GRB occurs when there

is a collision of two neutron stars. Regardless of being Short GRBs or Long GRBs, one thing was certain, these are the most powerful class of explosions that are known to occur in our universe.

The Brightest of the Gamma-ray burst was witnessed recently and